

From Governance Framework to Practice

Applying a doctoral research framework for responsible AI-assisted technical documentation to a 12-month, cross-functional documentation engagement.

PREPARED BY	PREPARED FOR	DATE	BASIS
Nathan B. Smith, D.C.S.	Hiring Manager, Collins Aerospace	July 2026	Smith (2024) Doctoral Dissertation, CTU

Executive Summary

This role asks for something most technical writing hires don't: not just documentation skill, but judgment about *how* AI tools like Copilot should be used to produce documentation that has to survive an audit. I spent two years researching exactly this problem — how small and mid-size defense-contracting organizations can adopt large language models for technical documentation without compromising accuracy, security, or compliance — and built a governance framework to answer it. This paper summarizes that research and proposes how I'd adapt it to this engagement's specific scope across Engineering, Quality, Operations, Supply Chain, and Support.

In short: I'm not proposing to invent a governance approach on the job. I've already built one, tested it against ten practitioners in this exact industry, and learned where it needs to flex for an enterprise environment rather than a small contractor. What follows is that framework, what my own research found still needed work, and a concrete first-90-days plan for putting it into practice here.

01 THE CHALLENGE

AI drafting tools are arriving in regulated technical-writing environments faster than governance for them is. The upside is real: faster first drafts, faster summarization of source material, less time spent on mechanical editing. The risk is just as real in an aerospace/defense context: an AI tool will confidently invent a torque value, a tolerance, or a procedural step it was never given, and if there's no defined review gate, that fabrication can ride straight into a released work instruction.

A role spanning Engineering, Quality, Operations, Supply Chain, and Support multiplies this risk surface: five functions, each with its own source data, its own SMEs, and its own tolerance for ambiguity. Without a shared governance model, AI adoption tends to happen unevenly — some teams draft everything by hand, others paste sensitive content into whatever tool is open, and no one can say with confidence which published documents had AI involvement or how that content was verified.

02 RESEARCH FOUNDATION

My 2024 doctoral dissertation, *Governance Frameworks for Large Language Models in Technical Documentation to Support Small Defense Contracting Firms* (Colorado Technical University), used a Design Science Research methodology: a comprehensive literature review paired with semi-structured interviews with ten technical communicators actively producing documentation — including IETMs — for DoD-facing organizations. The result was the **LLM Usage Governance Framework**, built around five components:

Component	What It Governs
Ethical Guidelines	Human oversight of AI-drafted content, disclosure of where AI was used, and active checks for bias or misleading phrasing before content reaches a technician or end user.
Data Security Protocols	What may and may not be entered into an AI tool, plus access control and integrity checks on anything the tool produces.
Quality Assurance Processes	Verification of AI-drafted content against released engineering data, a defined standards- compliance edit pass, and a documented review record.
Training & Education	Building AI literacy and prompt-engineering competence across the documentation team so review effort goes down, not up, as adoption grows.
Governance Structure	Named roles, a decision path for approving or escalating AI-assisted content, and an audit trail of what was generated, by whom, and how it was reviewed.

The interviews surfaced five consistent themes, in the participants' own priority order: the non-negotiable need for accuracy and reliability in defense documentation; data security and confidentiality as the unanimous top concern; ethical use and bias mitigation; a clear call for structured governance with defined roles rather than informal norms; and the value of ongoing training as both a risk-reduction and an efficiency measure.

03 WHAT MY OWN EVALUATION FOUND

I didn't just publish the framework and stop. I had the same ten practitioners evaluate it against five criteria — usability, applicability, efficacy, completeness, and flexibility — and the feedback is directly relevant to an enterprise engagement like this one:

FINDING	WHAT IT MEANS FOR THIS ROLE
Applicability	Reviewers rated the framework highly applicable to large programs but noted it needed lighter-weight, modular elements for smaller efforts. In a five-function engagement, that means one governance model with scalable checklists — not identical process overhead for a one-page work instruction and a full IETM.
Completeness	Data protection and privacy guidance needed to be more explicit. I'd build that directly into the Data Security Protocols component from day one here, rather than leaving it implicit.
Efficacy	The framework guided major decisions well but needed sharper detail on day-to-day operational monitoring. That becomes the metrics and audit-record piece in the plan below.
Flexibility	Participants wanted a built-in mechanism for regular updates as tools and regulations change — not a static policy document. I'd propose a quarterly review cadence rather than a "set it and forget it" governance doc.
Usability	Some sections were reported as too complex for daily use. The templates and prompt library proposed below exist specifically to make the governance model something a writer reaches for on a Tuesday afternoon, not a binder that sits on a shelf.

Why this matters for you: These aren't hypothetical gaps I'm guessing at — they're documented feedback from practitioners doing this work today. Coming in with that already known means less time spent discovering these limitations the hard way, and more time spent building around them.

04 PROPOSED APPLICATION TO THIS ENGAGEMENT

The framework's five components map directly onto the role's stated responsibilities:

ROLE RESPONSIBILITY	FRAMEWORK COMPONENT(S) APPLIED
Create/maintain procedures, work instructions, forms, templates, guidance documents	Quality Assurance Processes + Governance Structure: a standard template and review checklist per document type, so every deliverable carries the same verification steps regardless of which function requested it.
Use Copilot/AI tools with prompt engineering to accelerate drafting	Training & Education: a shared, versioned prompt library by document type, so drafting quality doesn't depend on one person's individual prompting skill.
Partner with cross-functional SMEs; ensure AI content meets quality/compliance standards	Ethical Guidelines + Quality Assurance: a defined SME verification gate against released data before any AI-assisted content is published, with sign-off logged.
Improve documentation governance via templates, style guides, KM, AI-driven process improvement	Governance Structure with the quarterly-review mechanism from Section 3: governance that's revisited on a cadence, not written once and left static.

05 A PRACTICAL FIRST 90 DAYS

DAYS 1-30 • ASSESS
Map current state before changing anything Inventory existing procedures, templates, and forms across the five functions. Identify where AI tools are already being used informally, and by whom, so the governance model fits real behavior rather than an assumed one.
DAYS 31-60 • PILOT
Build and test on one document type Stand up a template, a prompt library entry, and a review checklist for a single, representative document type. Run it with one cross-functional SME pairing before scaling, and adjust based on what the evaluation criteria in Section 3 predict will need simplifying.
DAYS 61-90 • SCALE & GOVERN
Roll out, train, and set the review cadence Extend the piloted template set to additional document types and functions. Deliver a short training session on the prompt library and review checklist. Establish the quarterly governance review going forward, with a lightweight audit-record format for tracking what's AI-assisted and how it was verified.

06 WHY THIS COMBINATION

Twenty-three years as a Navy Surface Warfare Officer taught me how documentation failures actually happen in operational environments. Fifteen-plus years writing S1000D and MIL-STD documentation for the RQ-4 Global Hawk, EMALS/Ford-class carriers, and commercial aircraft maintenance publications — including at Collins Aerospace itself — taught me what good documentation governance looks like in practice. And two years of doctoral research gave me an evidence-based framework for the specific problem this role now has to solve: how to bring AI into that process without breaking it. I'd rather bring a tested starting point to day one than build a governance model from scratch on the clock.

Source: Smith, N. B. (2024). *Governance Frameworks for Large Language Models in Technical Documentation to Support Small Defense Contracting Firms: A Design Science Research Study* [Doctoral dissertation, Colorado Technical University]. Framework components, evaluation criteria, and interview themes are drawn from that study; their application to this specific role and organization is proposed, not yet implemented.